

Box 44.1 Genetic and Genomic Testing

quence of small, rigid, polyaromatic chain (PAC) and the presence of long, flexible, aliphatic chain (FAC) or spacer. The presence of small rigid units in a PAC or spacer is thought to be important for the formation of a liquid crystalline phase. The presence of a long flexible spacer is thought to be important for the formation of a liquid crystalline phase. The presence of a long flexible spacer is thought to be important for the formation of a liquid crystalline phase. The presence of a long flexible spacer is thought to be important for the formation of a liquid crystalline phase.

Box 44.2 Clinical Assessment and Management of Patients With Alport Syndrome

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Box 44.2 Clinical Assessment and Management of Patients With Alport Syndrome (Continued)

transports 2 inhibition in *Alport syndrome*, but the magnitude of the benefit in terms of reduction in progression to kidney failure has not been established. However, the fact that patients with *Alport syndrome* were not excluded from large-scale trials of therapy [10–12] has provided justification for clinical trials of treatment in this population, suggesting that it is well tolerated in this group. [13] It is hoped that trials of this class of medication in children with kidney disease will be able to establish the efficacy and safety of this class of drug and efficacy in *Alport syndrome*.

As a multisystem disorder affecting hearing and the eyes, as well as the kidneys, *ALAD* and *AFAP* can give significant problems for affected children and young people and guidelines recommend annual childhood screening for hearing impairment from the age of 5 years and annual eye examinations from the age of 10 years with *AFAP*, since language and educational development can be impacted if hearing loss is unrecognized in childhood. Frequent hearing tests are recommended for children with *ALAD* as childhood screening programs for myopia in many countries, [14] but an awareness of the risk of keratoconus, which may manifest itself as a mild refractive error in childhood, is also important for children. Eye and ear complications are rare in people with a single autosomal pathogenic variant, and although hearing loss is more common in people with *AFAP*, it is not always detected (as required) should be sought in young people in this category with significant hearing impairment.

There is no accurate, best current treatment modality is kidney transplantation, but since many patients will have affected (or heterozygous) first-degree relatives, access to living-related transplantation is possible for some patients.

compared with non-Mendelian causes of kidney disease. As an important question relates to the suitability of heterozygous relatives as donors, it is important to know whether heterozygous carriers of *COL4A3* and *COL4A4* mutations are at risk of developing kidney disease. Guidelines suggest that other options should be explored and the pros and cons of predonation kidney biopsy should be discussed with the patient and family. The risk of developing kidney disease while living is not as empiric evidence allows quantification of the effect on risk of future kidney disease as in people with *AFAP* and *ALAD* [15] and the risk is greater than for people of similar health, age, and sex with that type I collagen genes, so decisions should be made to ensure that the patient is fully informed of the risks of not donating the organ. Data revealing whether allografts from heterozygotes fare less well are also lacking, further affecting the decision to donate. It is important to note that heterozygotes may have said that kidney donation by heterozygous females is not recommended, as the risk of developing kidney disease in having become pregnant in the future is generally considered to be high. The risk of developing kidney disease in heterozygotes is not as high as in people with *AFAP* and *ALAD* and single kidney during pregnancy are difficult to quantify.

A rare complication of kidney transplantation in patients with *COL4A3* and *COL4A4* mutations is glomerular disease directed against type IV collagen of the allograft glomerular basement membrane. With current immunosuppression regimens, this complication is rare, but it is important to be aware of it, with at least one sensitive variant. [16] While immunohistochemistry can show linear staining of the glomerular basement membrane, the use of immunofluorescence to detect anti-type IV antibodies are not thought to be sensitive in this situation unless the biotinic missing chain is collagen $\alpha 3(\text{IV})$, which contains

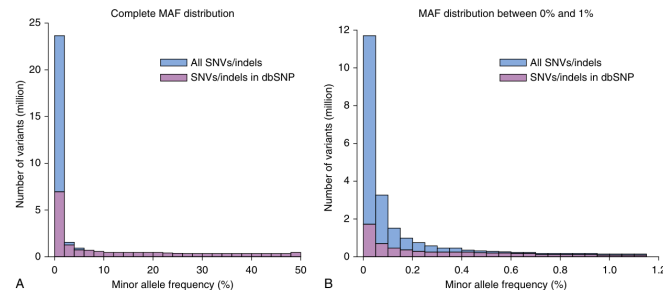


Fig. 44.1 The minor allele frequency (MAF) distribution in a European population. (Reproduced from Ameur A, Dahlberg J, Olason P, et al. SweGen: a whole-genome data resource of genetic variability in a cross-section of the Swedish population. *Eur J Hum Genet.* 2017;25(11):1253–1260.)

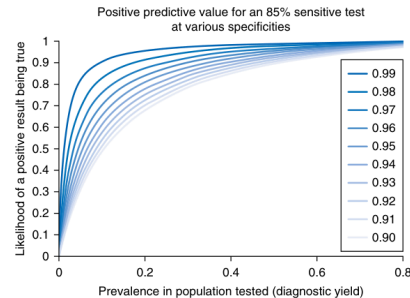


Fig. 44.2 Positive predictive value for an 85% sensitive test according to prior probability, or prevalence of the disease in the population in which the test is applied, with each line representing a specificity between 0.90 and 0.99.

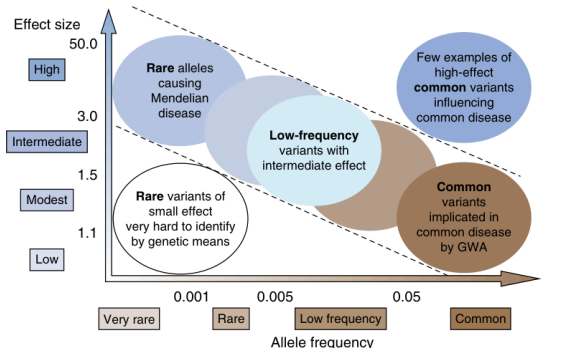


Fig. 44.4 Frequency and effect on disease risk of genetic variants, as perceived in 2010.¹⁴³



Fig. 44.3 Evidence framework for assigning attribution of pathogenicity to variants.⁴⁷ BP, Benign supporting; BS, benign strong; FS, pathogenic moderate; PP, pathogenic supporting; PS, pathogenic strong; PVS, pathogenic very strong.

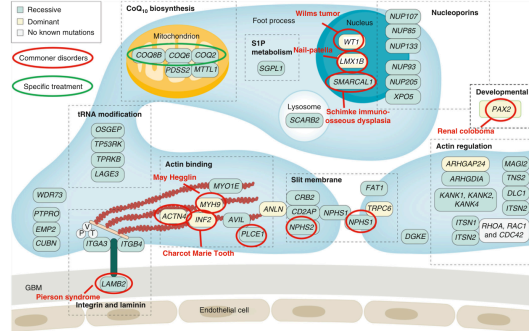


Fig. 44.5 Monogenic causes of podocytopathy. (Adapted from Kopp JB, Anders HJ, Susztak K, et al. Podocytopathies. *Nat Rev Dis Primers* 2020;6(1):68.)

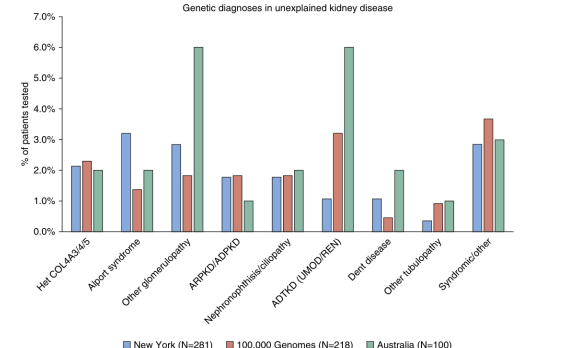


Fig. 44.6 Genetic diagnoses in patients undergoing genomic testing for unexplained kidney failure or nephropathy of unknown cause in New York,³ the UK 100,000 Genomes Project, and Australia.¹⁶⁴

Table 44.1 Variant Frequency and Typical Numbers in an Individual (data from 1000 Genomes and ExAC projects).^{145,146}

Frequency Category	Typical Number in an Individual
>5% (common)	300,000-500,000
0.5%-5% (uncommon, or low frequency)	100,000-200,000
0.01%-0.5% (rare)	50,000-100,000
<0.01% (very rare or private)	>1,000,000

Table_44_1

[illegible]

Table 44.4

Table 4.4.2. Rules for Combining Criteria to Classify Sequence Variants^a

Pathogenic
1. 1 Very Strong (PVS1) AND
a. ≥1 Strong (PS1-PM5) OR
b. ≥1 Moderate (PM1-PM6) OR
c. 1 Moderate (PM1-PM6) and 1 Supporting (PP1-PP5) OR
d. ≥2 Supporting (PP1-PP5)
2. ≥2 Strong (PS1-PS4) OR
3. 1 Strong (PS1-PS4) AND
a. ≥2 Moderate (PM1-PM6) OR
b. 1 Moderate (PM1-PM6) AND ≥2 Supporting (PP1-PP5) OR
c. 1 Moderate (PM1-PM6) AND ≥4 Supporting (PP1-PP5)
Likely Pathogenic
1. 1 Very Strong (PVS1) AND 1 Moderate (PM1-PM6) OR
2. 1 Strong (PS1-PS4) AND 1-2 Moderate (PM1-PM6) OR
3. 1 Strong (PS1-PS4) AND ≥2 Supporting (PP1-PP5) OR
4. 1 Moderate (PM1-PM6) AND ≥2 Supporting (PP1-PP5) OR
5. 2 Moderate (PM1-PM6) AND ≥2 Supporting (PP1-PP5) OR
6. 1 Moderate (PM1-PM6) AND ≥4 Supporting (PP1-PP5)
Benign
1. 1 Stand-Alone (BS1) OR
2. ≥2 Strong (BS1-BS4)
Likely Benign
1. 1 Strong (BS1-BS4) and 1 Supporting (BP1-BP7) OR
2. ≥2 Supporting (BP1-BP7)

^aVariants should be classified as Uncertain Significance if either variants are unmet or at least one variant for benign and pathogenic are contradictory.

Table_44_2

Table 4.4.5 Inherited Kidney Tumor Syndromes					
Syndrome	Gene	Mode of Inheritance	Renal and Histologic Features	Extrarenal Features	Notes
Von Hippel-Lindau (VHL) disease	VHL	AD	Clear cell RCC; renal cysts	CNS hemangioblastoma; retinal angiomas; pancreatic, renal, and spinal cysts; pheochromocytoma and other neuroendocrine tumors	Symptoms treated with VEGF inhibitor bevacizumab (not licensed)
Hereditary hemochromatosis and RCC (BAPCC)	FH	AD	Type 2 papillary RCC, renal cysts	Uveal melanoma; painful cutaneous telangiectases	Biallelic FH mutations associated with severe metabolic disease; HFE deficiency
Birt-Hogg-Darbo (BHD) disease	FLCN	AD	Chromophobe or papillary RCC; renal cysts	Pulmonary cysts and pneumothoraces; cutaneous trichodiscomas, fibrofolliculomas, collagen cylindromas	RCC occurs in <20% of those affected
Hereditary papillary renal cell carcinoma (RCCFH) syndrome	MET	AD	Renal cysts; Type 1 papillary RCC	Pneumothoraces; cutaneous trichodiscomas, fibrofolliculomas	Highly penetrant; acquired mutations
Prader-Willi syndrome	SNRNP40	AD	SNRNP-deficient RCC (nontranslocating)	Phenochromocytoma; Paraganglioma	SDNA is least common; bilateral SDNA
Paraganglioma	SDHB	AD	SDHB-deficient RCC (translocating)		SDNA is strong cause of Leigh syndrome
Pheochromocytoma	SDHA	AD			AMLs, a strong cause of mortality in adults with TSC2
Pheochromocytoma	SDHC	AD			causes developmental delay if rapidly extending AMLs
Tuberous sclerosis complex	TSC1	AD	Angiomyolipoma, renal cysts (rarely oncogenic) or aggressive epithelial AML	Facial angiofibromas, ungual fibromas, shagreen patch, hypopigmented macules, developmental delay, cardiac arrhythmia, subependymal giant cell cysts; renal cysts; lymphangioleiomyomatosis (usually in females); cardiac pharyngomas in infancy; renal hamartoma or childhood	Symptoms treated with mTOR inhibitor sirolimus (not licensed)
	TSC2	AD			
Coatsen syndrome	PTEN	AD	Multifocal RCC	Mucocystic IP; gonadal hamas, thyroid, endometrial, breast	

Table 44.5

Condition	Associated Locus (Represented by Nearest Gene)	Odds Ratio	Example Reference
IgA nephropathy	FCRL3, TNFSF4/18, CFH/CFHR1-3, REL, CD28, PF4V1, IRF4/ DUSP22, LY86, HLA-DRA, DQB, DQA, DPA, DQB, DEFA1/4, LYN, ANXA3, TNFSF8/15, CARD9, REEP3, ZMIZ1, OVOL1/ RELA, ETS1, IGH, ITGAM/ITGAX, IRF8, TNFSF12/13, TNFSF15/38, FCAR, LIF/OSM	1.12-1.33	26
IgA vasculitis	HLA-DQB, DRB	1.59	147
IC-MPGN/C3G	HLA-DQA1/DRB1	1.93	103
Membranous nephropathy	HLA-DQA2/DRB, PLA2R1, IRF4, NKFB1	1.25-2.41*	148
Steroid-sensitive nephrotic syndrome	HLA-DQB, NPS1, CALHM6, AHI1, TNFSF15, CLEC16A, CD28, BTHC	0.46-2.04	149
PR3 vasculitis	HLA-DP, SERPINA1, ARGHAP18, PRTN3	0.53-1.55	35
MPO-vasculitis	HLA-DQ	0.65	35
Eosinophilic granulomatosis with polyangiitis	HLA-DQB, RNU7-106P	1.62-2.0*	150
Posterior urethral valves	TBX5, PTK7	0.4-7.0	151
Bladder exstrophy	LPNH2, EFNA1, SLC50A1, DPM3, KRTPAC2, ISL1, TRIM29, SYT1, PAWR, GOSR2, LINC01974, LINC01716, HMG1P47, ISL1-DT	1.47-2.35	152
Urinary stone disease	ALPL, GOKR, DGKD, ABCG2, SCL34A1, KKNK5, SLC22A2, HIBADH, AQP1, POU2AF, DGKH, WDR72, UMOD, SCNN1B, BCAS, SOX9, GIPCI, CYP24A1, CLDN14, BCR	1.05-1.24	27

*Strong evidence of epistasis between PLA2R1 and HLA-DRB (see text).